

Report Period	2026-08-28 - 2026-09-04
Generated At	2026-09-04 13:24:27
Classification	ML: Isolation Forest + Random Forest

1. Executive Summary

Metric	Value	Traffic Summary	Count
Report Period	2026-08-28 - 2026-09-04	Total Flows Observed	118619
Total Threats Mitigated	67	Total Packets Analyzed	24064245463
High Priority	23	Malicious Packets	24046402426
Low Priority	44	True Negatives Passed	91514
Manual Releases	0		
Manual Blocks	0		
ML Metrics	Value	Attack Vector	Count
False Positives	96	ICMP Flood	13
FP Rate	0.11%	SYN Flood	17
IF Precision	99.64%	UDP Flood	20
IF Recall	93.00%	Uncertain	17
IF F1-Score	96.21%		
IF Accuracy	98.21%	Action	Count
		Quarantined	13
		Time Ban	10
		Blackhole	0
		Released	14

2. Performance Benchmark

2a. Isolation Forest-Anomaly Detection

Metric	Value	Description
Precision	99.64%	Share of flagged anomalies that were genuine attacks
Recall (TPR)	93.00%	Share of actual attacks successfully flagged
F1-Score	96.21%	Balanced measure combining Precision and Recall
Accuracy	98.21%	Overall proportion of correct anomaly decisions
False Positive Rate (FPR)	0.11%	Normal traffic incorrectly flagged as an attack
False Negative Rate (FNR)	7.00%	Actual attacks that went undetected
True Positive Rate (TPR)	93.00%	Same measure as Recall, attacks correctly flagged
True Negative Rate (TNR)	99.89%	Normal traffic correctly identified as safe

	Predicted: Attack	Predicted: Normal
Actual: Attack	TP 26943	FN 2029
Actual: Normal	FP 96	TN 89485

2b. Random Forest-Attack Classification

Metric	Value	Description
Precision	100.00%	Share of classified flows assigned the correct attack type
Recall	100.00%	Share of attacks of each type that were correctly identified
F1-Score	100.00%	Balanced measure combining Precision and Recall
Accuracy	100.00%	Overall proportion of correct classifications

	Predicted: SYN	Predicted: ICMP	Predicted: UDP
Act: SYN	13989	0	0
Act: ICMP	0	9446	0
Act: UDP	0	0	3508

2c. Response Latency

Metric	Value	Description
Detection Time	338.0 ms	Average time taken to detect an attack
Mitigation Response Time	1.1 ms	Interval between the anomaly flag and the FlowMod blocking rule install

2d. Controller Resource Overhead

Metric	Value	Description
Controller CPU (Baseline)	1.28%	Ryu controller CPU usage under normal traffic
Controller CPU (Active Attack)	2.61%	Ryu controller average CPU usage during a DDoS simulation
Controller CPU (Mitigation)	2.25%	Ryu controller CPU usage during simultaneous detection and mitigation

3. Offences Summary

Source IP	Sessions	Max Offences	First Seen	Last Seen
10.0.0.6	5	5	2026-09-04 12:59:48	2026-09-04 13:13:59
10.0.0.8	5	5	2026-09-04 12:59:48	2026-09-04 13:23:33
10.0.0.17	4	4	2026-09-04 13:02:33	2026-09-04 13:05:45
10.0.0.24	4	4	2026-09-04 13:12:29	2026-09-04 13:15:56
10.0.0.9	4	4	2026-09-04 13:09:52	2026-09-04 13:22:28
10.0.0.16	3	3	2026-09-04 13:02:26	2026-09-04 13:05:32
10.0.0.18	3	3	2026-09-04 13:02:30	2026-09-04 13:05:36
10.0.0.19	3	3	2026-09-04 13:02:27	2026-09-04 13:05:33
10.0.0.20	3	3	2026-09-04 13:22:26	2026-09-04 13:23:52
10.0.0.21	3	3	2026-09-04 13:22:27	2026-09-04 13:23:54
10.0.0.22	3	3	2026-09-04 13:22:28	2026-09-04 13:23:54
10.0.0.23	3	3	2026-09-04 13:12:26	2026-09-04 13:15:47
10.0.0.25	3	3	2026-09-04 13:12:27	2026-09-04 13:15:48
10.0.0.1	1	1	2026-09-04 13:09:47	2026-09-04 13:09:57
10.0.0.10	1	1	2026-09-04 12:59:45	2026-09-04 12:59:56
10.0.0.2	1	1	2026-09-04 13:19:48	2026-09-04 13:19:58
10.0.0.3	1	1	2026-09-04 13:09:46	2026-09-04 13:09:56
10.0.0.5	1	1	2026-09-04 12:59:45	2026-09-04 12:59:56
10.0.0.7	1	1	2026-09-04 12:59:46	2026-09-04 12:59:56

4. Chronological Mitigation Log

Timestamp	Source IP	Class	Vector	Confidence	Priority	Action
2026-09-04 12:59:48	10.0.0.6	DDoS	UDP Flood	90.6%	High	Time Ban (1m)
2026-09-04 12:59:48	10.0.0.8	DDoS	UDP Flood	90.9%	High	Time Ban (1m)
2026-09-04 12:59:56	10.0.0.5	DDoS	Uncertain	59.6%	Low	Released
2026-09-04 12:59:56	10.0.0.10	DDoS	Uncertain	68.7%	Low	Released
2026-09-04 12:59:56	10.0.0.7	DDoS	Uncertain	68.4%	Low	Released
2026-09-04 13:00:49	10.0.0.6	DDoS	UDP Flood	90.9%	High	Released
2026-09-04 13:00:49	10.0.0.8	DDoS	UDP Flood	90.9%	High	Released
2026-09-04 13:01:11	10.0.0.8	DDoS	UDP Flood	90.9%	High	Quarantined
2026-09-04 13:02:26	10.0.0.16	DDoS	Uncertain	-	Low	Quarantined
2026-09-04 13:02:27	10.0.0.19	DDoS	Uncertain	-	Low	Quarantined
2026-09-04 13:02:30	10.0.0.18	DDoS	Uncertain	-	Low	Quarantined
2026-09-04 13:02:32	10.0.0.16	DDoS	SYN Flood	93.1%	Critical	Time Ban (1m)
2026-09-04 13:02:33	10.0.0.16	DDoS	SYN Flood	93.1%	Critical	Time Ban
2026-09-04 13:02:33	10.0.0.19	DDoS	SYN Flood	90.2%	Critical	Time Ban (1m)
2026-09-04 13:02:33	10.0.0.17	DDoS	Uncertain	-	Low	Quarantined
2026-09-04 13:02:33	10.0.0.19	DDoS	SYN Flood	91.0%	Critical	Time Ban
2026-09-04 13:02:36	10.0.0.18	DDoS	SYN Flood	93.1%	Critical	Time Ban (1m)
2026-09-04 13:02:36	10.0.0.18	DDoS	SYN Flood	93.1%	Critical	Time Ban
2026-09-04 13:02:39	10.0.0.17	DDoS	SYN Flood	92.7%	Critical	Time Ban (1m)
2026-09-04 13:02:39	10.0.0.17	DDoS	SYN Flood	92.7%	Critical	Time Ban
2026-09-04 13:03:32	10.0.0.16	DDoS	SYN Flood	93.3%	Critical	Time Ban (2m)
2026-09-04 13:03:33	10.0.0.19	DDoS	SYN Flood	94.0%	Critical	Time Ban (2m)
2026-09-04 13:03:36	10.0.0.18	DDoS	SYN Flood	93.1%	Critical	Time Ban (2m)
2026-09-04 13:03:39	10.0.0.17	DDoS	SYN Flood	94.6%	Critical	Released
2026-09-04 13:03:45	10.0.0.17	DDoS	SYN Flood	94.6%	Critical	Time Ban (2m)
2026-09-04 13:05:32	10.0.0.16	DDoS	SYN Flood	93.3%	Critical	Time Ban (5m)
2026-09-04 13:05:33	10.0.0.19	DDoS	SYN Flood	94.0%	Critical	Time Ban (5m)
2026-09-04 13:05:36	10.0.0.18	DDoS	SYN Flood	93.1%	Critical	Time Ban (5m)
2026-09-04 13:05:45	10.0.0.17	DDoS	SYN Flood	94.6%	Critical	Time Ban (5m)
2026-09-04 13:09:52	10.0.0.9	DDoS	UDP Flood	92.4%	High	Time Ban (1m)
2026-09-04 13:09:56	10.0.0.3	DDoS	Uncertain	59.1%	Low	Released
2026-09-04 13:09:57	10.0.0.1	DDoS	Uncertain	58.6%	Low	Released
2026-09-04 13:10:53	10.0.0.9	DDoS	UDP Flood	92.4%	High	Released
2026-09-04 13:11:25	10.0.0.6	DDoS	UDP Flood	73.6%	High	Quarantined
2026-09-04 13:12:26	10.0.0.23	DDoS	Uncertain	-	Low	Quarantined
2026-09-04 13:12:27	10.0.0.25	DDoS	Uncertain	-	Low	Quarantined
2026-09-04 13:12:29	10.0.0.24	DDoS	Uncertain	-	Low	Quarantined

Timestamp	Source IP	Class	Vector	Confidence	Priority	Action
2026-09-04 13:12:47	10.0.0.23	DDoS	ICMP Flood	92.0%	High	Time Ban (1m)
2026-09-04 13:12:48	10.0.0.25	DDoS	ICMP Flood	91.9%	High	Time Ban (1m)
2026-09-04 13:12:48	10.0.0.23	DDoS	ICMP Flood	92.0%	Critical	Time Ban
2026-09-04 13:12:48	10.0.0.25	DDoS	ICMP Flood	91.9%	Critical	Time Ban
2026-09-04 13:12:50	10.0.0.24	DDoS	ICMP Flood	92.0%	High	Time Ban (1m)
2026-09-04 13:12:50	10.0.0.24	DDoS	ICMP Flood	92.0%	Critical	Time Ban
2026-09-04 13:13:47	10.0.0.23	DDoS	ICMP Flood	92.3%	High	Time Ban (2m)
2026-09-04 13:13:48	10.0.0.25	DDoS	ICMP Flood	92.3%	High	Time Ban (2m)
2026-09-04 13:13:50	10.0.0.24	DDoS	ICMP Flood	92.2%	High	Released
2026-09-04 13:13:56	10.0.0.24	DDoS	ICMP Flood	92.2%	Critical	Time Ban (2m)
2026-09-04 13:15:47	10.0.0.23	DDoS	ICMP Flood	96.0%	Critical	Time Ban (5m)
2026-09-04 13:15:48	10.0.0.25	DDoS	ICMP Flood	96.0%	Critical	Time Ban (5m)
2026-09-04 13:15:56	10.0.0.24	DDoS	ICMP Flood	96.0%	Critical	Time Ban (5m)
2026-09-04 13:19:58	10.0.0.2	DDoS	Uncertain	58.9%	Low	Released
2026-09-04 13:21:26	10.0.0.9	DDoS	Uncertain	48.4%	Low	Quarantined
2026-09-04 13:22:26	10.0.0.20	DDoS	Uncertain	-	Low	Quarantined
2026-09-04 13:22:27	10.0.0.21	DDoS	Uncertain	-	Low	Quarantined
2026-09-04 13:22:28	10.0.0.22	DDoS	Uncertain	-	Low	Quarantined
2026-09-04 13:22:46	10.0.0.20	DDoS	UDP Flood	76.0%	High	Time Ban (1m)
2026-09-04 13:22:46	10.0.0.20	DDoS	UDP Flood	76.0%	High	Time Ban
2026-09-04 13:22:47	10.0.0.21	DDoS	UDP Flood	78.0%	High	Time Ban (1m)
2026-09-04 13:22:48	10.0.0.21	DDoS	UDP Flood	78.0%	High	Time Ban
2026-09-04 13:22:48	10.0.0.22	DDoS	UDP Flood	78.3%	High	Time Ban (1m)
2026-09-04 13:22:49	10.0.0.22	DDoS	UDP Flood	78.3%	High	Time Ban
2026-09-04 13:23:46	10.0.0.20	DDoS	UDP Flood	86.9%	High	Released
2026-09-04 13:23:47	10.0.0.21	DDoS	UDP Flood	78.0%	High	Released
2026-09-04 13:23:48	10.0.0.22	DDoS	UDP Flood	85.2%	High	Released
2026-09-04 13:23:52	10.0.0.20	DDoS	UDP Flood	86.9%	Critical	Time Ban (2m)
2026-09-04 13:23:54	10.0.0.22	DDoS	UDP Flood	85.2%	Critical	Time Ban (2m)
2026-09-04 13:23:54	10.0.0.21	DDoS	UDP Flood	79.7%	Critical	Time Ban (2m)

5. IP Attack History (Completed Sessions)

Source IP	Vector	IF Score	Conf	Priority	Phase	Offences	Duration	Unblocked At	Reason
10.0.0.22	UDP Flood	0.6250	85.2%	Critical	Phase 2	3	6s	2026-09-04 13:23:54	Attack persisted - escalating to time ban
10.0.0.21	UDP Flood	0.6195	79.7%	Critical	Phase 2	3	5s	2026-09-04 13:23:54	Attack persisted - escalating to time ban

Source IP	Vector	IF Score	Conf	Priority	Phase	Offences	Duration	Unblocked At	Reason
10.0.0.20	UDP Flood	0.6260	86.9%	Critical	Phase 2	3	5s	2026-09-04 13:23:52	Attack persisted - escalating to time ban
10.0.0.22	UDP Flood	0.6215	85.2%	High	Phase 2	1	1m 20s	2026-09-04 13:23:48	Ban Expired
10.0.0.21	UDP Flood	0.5831	78.0%	High	Phase 2	1	1m 20s	2026-09-04 13:23:47	Ban Expired
10.0.0.20	UDP Flood	0.5992	86.9%	High	Phase 2	1	1m 20s	2026-09-04 13:23:46	Ban Expired
10.0.0.8	Uncertain	0.5916	68.7%	Medium	Phase 1	5	15s	2026-09-04 13:23:33	Attack Stopped
10.0.0.22	UDP Flood	0.6345	78.3%	High	Phase 2	1	20s	2026-09-04 13:22:48	Attack persisted - escalating to time ban
10.0.0.21	UDP Flood	0.6195	78.0%	High	Phase 2	1	20s	2026-09-04 13:22:47	Attack persisted - escalating to time ban
10.0.0.20	UDP Flood	0.6334	76.0%	High	Phase 2	1	20s	2026-09-04 13:22:46	Attack persisted - escalating to time ban
10.0.0.9	Uncertain	0.6298	68.7%	Medium	Phase 1	4	16s	2026-09-04 13:22:28	Attack Stopped
10.0.0.9	Uncertain	0.6172	68.7%	Medium	Phase 1	3	16s	2026-09-04 13:22:02	Attack Stopped
10.0.0.9	Uncertain	0.6293	48.4%	Low	Phase 1	2	10s	2026-09-04 13:21:36	Attack Stopped
10.0.0.2	Uncertain	0.6091	58.9%	Low	Phase 1	1	10s	2026-09-04 13:19:58	Attack Stopped
10.0.0.24	ICMP Flood	0.6397	96.0%	Critical	Phase 2	4	2m 6s	2026-09-04 13:15:56	Attack persisted - escalating to time ban
10.0.0.25	ICMP Flood	0.6358	96.0%	Critical	Phase 2	3	3m 21s	2026-09-04 13:15:48	Attack persisted - escalating to time ban
10.0.0.23	ICMP Flood	0.6354	96.0%	Critical	Phase 2	3	3m 21s	2026-09-04 13:15:47	Attack persisted - escalating to time ban
10.0.0.6	UDP Flood	0.6283	92.5%	Critical	Phase 1	5	6s	2026-09-04 13:13:59	Attack Stopped
10.0.0.24	ICMP Flood	0.6419	92.2%	Critical	Phase 2	3	6s	2026-09-04 13:13:56	Attack persisted - escalating to time ban
10.0.0.24	ICMP Flood	0.6410	92.2%	High	Phase 2	1	1m 21s	2026-09-04 13:13:50	Ban Expired
10.0.0.25	ICMP Flood	0.6378	92.3%	High	Phase 2	2	1m 21s	2026-09-04 13:13:48	Attack persisted - escalating to time ban
10.0.0.23	ICMP Flood	0.5960	92.3%	High	Phase 2	2	1m 21s	2026-09-04 13:13:47	Attack persisted - escalating to time ban
10.0.0.6	UDP Flood	0.5889	92.5%	High	Phase 1	4	21s	2026-09-04 13:13:25	Attack Stopped
10.0.0.8	UDP Flood	0.6222	92.4%	Critical	Phase 1	4	6s	2026-09-04 13:12:55	Attack Stopped

Source IP	Vector	IF Score	Conf	Priority	Phase	Offences	Duration	Unblocked At	Reason
10.0.0.24	ICMP Flood	0.6419	92.0%	High	Phase 2	1	21s	2026-09-04 13:12:50	Attack persisted - escalating to time ban
10.0.0.25	ICMP Flood	0.6414	91.9%	High	Phase 2	1	21s	2026-09-04 13:12:48	Attack persisted - escalating to time ban
10.0.0.23	ICMP Flood	0.6419	92.0%	High	Phase 2	1	21s	2026-09-04 13:12:47	Attack persisted - escalating to time ban
10.0.0.6	UDP Flood	0.6168	92.4%	High	Phase 1	3	20s	2026-09-04 13:12:38	Attack Stopped
10.0.0.8	UDP Flood	0.5822	92.4%	High	Phase 1	3	20s	2026-09-04 13:12:22	Attack Stopped
10.0.0.6	UDP Flood	0.6102	73.9%	High	Phase 1	2	21s	2026-09-04 13:11:46	Attack Stopped
10.0.0.9	UDP Flood	0.5591	92.4%	High	Phase 2	1	1m 1s	2026-09-04 13:10:53	Ban Expired
10.0.0.1	Uncertain	0.6107	58.6%	Low	Phase 1	1	10s	2026-09-04 13:09:57	Attack Stopped
10.0.0.3	Uncertain	0.6239	59.1%	Low	Phase 1	1	10s	2026-09-04 13:09:56	Attack Stopped
10.0.0.17	SYN Flood	0.6242	94.6%	Critical	Phase 2	4	2m 5s	2026-09-04 13:05:45	Attack persisted - escalating to time ban
10.0.0.18	SYN Flood	0.6142	93.1%	Critical	Phase 2	3	3m 6s	2026-09-04 13:05:36	Attack persisted - escalating to time ban
10.0.0.19	SYN Flood	0.6325	94.0%	Critical	Phase 2	3	3m 6s	2026-09-04 13:05:33	Attack persisted - escalating to time ban
10.0.0.16	SYN Flood	0.6218	93.3%	Critical	Phase 2	3	3m 6s	2026-09-04 13:05:32	Attack persisted - escalating to time ban
10.0.0.17	SYN Flood	0.6203	94.6%	Critical	Phase 2	3	5s	2026-09-04 13:03:45	Attack persisted - escalating to time ban
10.0.0.17	SYN Flood	0.6175	94.6%	Critical	Phase 2	1	1m 6s	2026-09-04 13:03:39	Ban Expired
10.0.0.18	SYN Flood	0.6317	93.1%	Critical	Phase 2	2	1m 6s	2026-09-04 13:03:36	Attack persisted - escalating to time ban
10.0.0.19	SYN Flood	0.6402	94.0%	Critical	Phase 2	2	1m 6s	2026-09-04 13:03:33	Attack persisted - escalating to time ban
10.0.0.16	SYN Flood	0.6245	93.3%	Critical	Phase 2	2	1m 6s	2026-09-04 13:03:32	Attack persisted - escalating to time ban
10.0.0.17	SYN Flood	0.6282	92.7%	Critical	Phase 2	1	6s	2026-09-04 13:02:39	Attack persisted - escalating to time ban
10.0.0.18	SYN Flood	0.6422	93.1%	Critical	Phase 2	1	6s	2026-09-04 13:02:36	Attack persisted - escalating to time ban

Source IP	Vector	IF Score	Conf	Priority	Phase	Offences	Duration	Unblocked At	Reason
10.0.0.19	SYN Flood	0.6525	90.2%	Critical	Phase 2	1	6s	2026-09-04 13:02:33	Attack persisted - escalating to time ban
10.0.0.16	SYN Flood	0.6360	93.1%	Critical	Phase 2	1	6s	2026-09-04 13:02:32	Attack persisted - escalating to time ban
10.0.0.8	UDP Flood	0.5811	90.9%	High	Phase 1	2	21s	2026-09-04 13:01:32	Attack Stopped
10.0.0.6	UDP Flood	0.5836	90.9%	High	Phase 2	1	1m 1s	2026-09-04 13:00:49	Ban Expired
10.0.0.8	UDP Flood	0.5791	90.9%	High	Phase 2	1	1m 1s	2026-09-04 13:00:49	Ban Expired
10.0.0.5	Uncertain	0.6137	59.6%	Low	Phase 1	1	11s	2026-09-04 12:59:56	Attack Stopped
10.0.0.10	Uncertain	0.6182	68.7%	Low	Phase 1	1	11s	2026-09-04 12:59:56	Attack Stopped
10.0.0.7	Uncertain	0.6112	68.4%	Low	Phase 1	1	10s	2026-09-04 12:59:56	Attack Stopped

Verification and Approval

I hereby verify that the findings in this incident report are accurate and that the listed mitigation actions were carried out under my supervision.

SIGNATURE OVER PRINTED NAME

Network Administrator

DATE